BATTERY INPUTS / SERIES LINK + RESERVOIR

Series: J4 upper battery, J30 lower battery; pin 1 +, pin 2 −.

Single 24 V PSU: + to J4.1, − to J30.2; leave J4.2 / J30.1 unused.

BAT_MID_12V joins J4.2 to J30.1 only; it is not board ground.

USB-C / PROGRAMMING ONLY

PROGRAMMING ONLY — power the controller from its normal supply.

USB VBUS is isolated from VIN_5V. No USB-PD or USB-UART bridge.

VBUS presence: J1.20 / ESP GPIO8; firmware must monitor this input.

A6/B6 = D+; A7/B7 = D−. Each CC pin has its own 5.1k pulldown.

LOCAL POWER + BRIDGE LEG INDICATORS

Each bridge lamp senses its switching node through 12k; no FPGA status pins.

1N4148W protects each LED from negative switching-node excursions.

Brightness follows positive duty cycle; not a fault, enable or current monitor.

GDT 1 / FPGA COMMAND DAMPING

FPGA owns enable, zero-current shutdown, deadtime and flux balance.

DESIGN NOTES

Input wiring: see BATTERY INPUTS / SERIES LINK + RESERVOIR.

Values and power-stage thermal performance require validation.

24 V TO 12 V / DRIVER BIAS

LMZ14202H: shielded inductor inside the module.

UVLO: approx. 21.0 V on / 19.4 V off; soft start 0.47 ms.

Output bank: verify DC-biased capacitance before release.

2 A device rating; thermal / load qualification pending.

24 V TO 5 V / LOGIC

LMZ14202H: shielded inductor inside the module.

UVLO: approx. 21.0 V on / 19.4 V off; soft start 0.47 ms.

Output bank: verify DC-biased capacitance before release.

2 A device rating; thermal / load qualification pending.

AUX 1 / UNITY BUFFER

1k / 2.2nF: 72 kHz nominal; ADC-side 100R / 1nF follows.

1M DC load; shared return; powered operation only.

ADC / AD7606C-16

OS[2:0]=111: software mode

Configure ranges and high bandwidth before capture.

ANALOG INPUTS / ADC RC FILTERS

CH1-4: AUX1, AUX2, AUX3, AUX4 / 0-5 V; CH5/6 CST 1; CH7/8 CST 2

100R / 1nF: EMI filtering, not anti-alias filtering.

ADC / LOCAL BYPASS + REFERENCE

REGCAP_A and REGCAP_D are internal-regulator outputs: bypass only.

REFCAP pins 44/45 share C9; REF pin 42 uses C8.

GDT 2 / FULL BRIDGE + GATE BIAS

Precharge: both lows ON, both highs OFF. Insert deadtime before each leg commutation.

UD2.7: 2.2uF / 50V X7R DC block, with parallel 15R + 1uF / 50V X7R damping branch.

CST 1 / INPUT + PRECISION RECTIFIER

10R burden: 2 x 20R / nominal +/-2 V. Choose CT ratio N = 5 x primary peak amperes.

Vhalf = −max(Vct,0); U41 sums −(Vct + 2 Vhalf) = |Vct|. 1k/10nF: 10 us smoothing, not peak hold.

Keep a secondary clamp at the CT itself if the cable can unplug. Overcurrent decisions are made by the FPGA from ADC samples.

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SHARED −5V / CHARGE PUMP

250 kHz charge pump; 10R/10uF ripple filter.

Negative rail feeds both rectifiers and the AUX buffers.

CST 1 / PHASE COMPARATOR + REFERENCE

Vphase = 1.65 V + Vct/2; compare with CT1_MID.

22pF EMI filter; 6mV internal comparator hysteresis.

Timing threshold + delay must be calibrated versus amplitude.

Phase midpoint: 1.65 V. Wait >=100 ms after rails stabilize before arming.

Sheet: /
File: controller.kicad_sch

Title: Tesla coil controller / sensing, control and power

Size: User | Date: | **Rev: A0 — NOT FOR FABRICAT**
KiCad E.D.A. 10.0.6 | Id: 1/1